**Math 510** **Fall 2026**
**Homework 4 Solutions** **Drew Armstrong**

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**1. Trivial Kernel.**

(a) Prove that a linear function $f$ is injective if and only if $ker(f) = \{\mathbf{0}\}$.
(b) Prove that the columns of a matrix $A$ are independent if and only if $\mathcal{N}(A) = \{\mathbf{0}\}$.

(a): Consider any linear function $f : U \to V$. First suppose that $f$ is injective. To show that $ker(f) = \{\mathbf{0}\}$, consider any element $\mathbf{u} \in ker(f)$, so that $f(\mathbf{u}) = \mathbf{0}$. Since $f$ is linear, we also have $f(\mathbf{0}) = \mathbf{0}$. Then since $f(\mathbf{u}) = f(\mathbf{0})$, the injectivity of $f$ implies $\mathbf{u} = \mathbf{0}$. For the other direction, suppose that $ker(f) = \{\mathbf{0}\}$. To show that $f$ is injective, suppose that $f(\mathbf{u}_1) = f(\mathbf{u}_2)$. Then we have

$$\begin{aligned} f(\mathbf{u}_1) &= f(\mathbf{u}_2) \\ f(\mathbf{u}_1) - f(\mathbf{u}_2) &= \mathbf{0} \\ f(\mathbf{u}_1 - \mathbf{u}_2) &= \mathbf{0} && \text{linearity of } f \\ \mathbf{u}_1 - \mathbf{u}_2 &= \mathbf{0} && ker(f) = \{\mathbf{0}\} \\ \mathbf{u}_1 &= \mathbf{u}_2. \end{aligned}$$

Hence $f$ is injective. □

(b): Consider any matrix $A \in \mathbb{C}^{m\times n}$ and let $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{C}^m$ be the column vectors of $A$. Let $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{C}^n$ be an arbitrary vector. We note that

$$\mathbf{x} \in \mathcal{N}(A) \iff A\mathbf{x} = \mathbf{0} \iff x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}.$$

First suppose that $\mathcal{N}(A) = \{\mathbf{0}\}$. Then we have

$$\begin{aligned} x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0} &\implies \mathbf{x} \in \mathcal{N}(A) \\ &\implies \mathbf{x} = \mathbf{0} \\ &\implies x_1 = x_2 = \cdots = x_n = 0. \end{aligned}$$

Conversely, suppose that $\mathbf{a}_1, \dots, \mathbf{a}_n$ are independent. Then

$$\begin{aligned} \mathbf{x} \in \mathcal{N}(A) &\implies x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0} \\ &\implies x_1 = x_2 = \cdots = x_n = 0 \\ &\implies \mathbf{x} = \mathbf{0}. \end{aligned}$$

□

**2. Rank-Nullity Theorem.** Let $U$ and $V$ be finite-dimensional vector spaces, and let $f : U \to V$ be a linear function. Prove that

$$\dim im(f) + \dim ker(f) = \dim U.$$

[Hint: Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be a basis for $ker(f)$, and extend this to a basis $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ for $U$. Prove that $f(\mathbf{u}_{k+1}), \dots, f(\mathbf{u}_n)$ is a basis for $im(f)$.]

Let $\dim U = n$ and $\dim ker(f) = k \leq n$. Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be a basis for $ker(f)$ and extend this to a basis $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ for $U$. We will show that $f(\mathbf{u}_{k+1}), \dots, f(\mathbf{u}_n)$ is a basis for $im(f)$, and hence

$$\dim im(f) = n - k = \dim U - \dim ker(f).$$

To show that it is a spanning set, consider any $\mathbf{v} \in im(f)$. By definition of the image, we have $\mathbf{v} = f(\mathbf{u})$ for some $\mathbf{u} \in U$. Write $\mathbf{u} = \alpha_1\mathbf{u}_1 + \cdots + \alpha_n\mathbf{u}_n$. Then by linearity of $f$ we have

$$\begin{aligned}\mathbf{v} &= f(\mathbf{u}) \\ &= \alpha_1 f(\mathbf{u}_1) + \cdots + \alpha_n f(\mathbf{u}_n) \\ &= \alpha_1\mathbf{0} + \cdots + \alpha_k\mathbf{0} + \alpha_{k+1}f(\mathbf{u}_{k+1}) + \cdots + \alpha_n f(\mathbf{u}_n) \\ &= \alpha_{k+1}f(\mathbf{u}_{k+1}) + \cdots + \alpha_n f(\mathbf{u}_n),\end{aligned}$$

and we conclude that $\mathbf{v}$ is in the span of $f(\mathbf{u}_{k+1}), \ldots, f(\mathbf{u}_n)$. To show that the set is independent, suppose that we have

$$\alpha_{k+1}f(\mathbf{u}_{k+1}) + \cdots + \alpha_n f(\mathbf{u}_n) = \mathbf{0}$$

for some scalars $\alpha_{k+1}, \ldots, \alpha_n$. By linearity, this implies

$$f(\alpha_{k+1}\mathbf{u}_{k+1} + \cdots + \alpha_n\mathbf{u}_n) = \alpha_{k+1}f(\mathbf{u}_{k+1}) + \cdots + \alpha_n f(\mathbf{u}_n) = \mathbf{0},$$

and hence $\alpha_{k+1}\mathbf{u}_{k+1} + \cdots + \alpha_n\mathbf{u}_n \in ker(f)$. Since $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is a basis for $ker(f)$ we can write

$$\alpha_{k+1}\mathbf{u}_{k+1} + \cdots + \alpha_n\mathbf{u}_n = \beta_1\mathbf{u}_1 + \cdots + \beta_k\mathbf{u}_k$$

for some scalars $\beta_1, \ldots, \beta_k$. But then we have

$$-\beta_1\mathbf{u}_1 - \cdots - \beta_k\mathbf{u}_k + \alpha_{k+1}\mathbf{u}_{k+1} + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}.$$

Finally, the independence of the vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ implies that all of the coefficients are zero. In particular, $\alpha_{k+1} = \cdots = \alpha_n = 0$. □

**3. Intersection of Linear Hyperplanes.** A *linear hyperplane* is a vector subspace $H \subseteq V$ of dimension $\dim V - 1$. Suppose that $V$ is also an inner product space.

(a) Prove that every linear hyperplane $H \subseteq V$ has the form

$$H = \mathbf{u}^\perp = \{\mathbf{x} \in V : \langle \mathbf{u}, \mathbf{x}\rangle = 0\},$$

for some unit vector $\mathbf{u} \in V$. [Hint: Consider the orthogonal complement $H^\perp \subseteq V$.]

(b) Given linear hyperplanes $H_i = \mathbf{u}_i^\perp$, prove that

$$H_1 \cap H_2 \cap \cdots \cap H_m = \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}^\perp.$$

(c) Express part (b) in terms of the nullspace and the row space of some matrix.

(a): We know that $\dim H + \dim H^\perp = \dim V$. Since $\dim H = \dim V - 1$, we must have $\dim H^\perp = 1$. This implies that $H^\perp = \text{Span}\{\mathbf{u}\}$ for some unit vector $\mathbf{u} \in V$. But then

$$H = (H^\perp)^\perp = \text{Span}\{\mathbf{u}\}^\perp = \mathbf{u}^\perp.$$

(b): We have

$$\begin{aligned}\mathbf{x} \in H_1 \cap H_2 \cap \cdots \cap H_m \quad &\iff \quad \mathbf{x} \in H_i \text{ for all } i \\ &\iff \quad \langle \mathbf{u}_i, \mathbf{x}\rangle = 0 \text{ for all } i \\ &\iff \quad \left\langle \sum \alpha_i\mathbf{u}_i, \mathbf{x} \right\rangle = 0 \text{ for all } \alpha_i \\ &\iff \quad \mathbf{x} \in \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}^\perp.\end{aligned}$$

(c): Let $\mathbf{u}_1, \ldots, \mathbf{u}_m$ be the columns of $U^*$, so that $\mathcal{R}(U) = \mathcal{C}(U^*) = \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$. Then since $\mathcal{N}(U) = \mathcal{R}(U)^\perp$, we conclude from part (b) that

$$\mathcal{N}(U) = H_1 \cap H_2 \cap \cdots \cap H_m.$$

In fact, we could have proved part (b) this way, because the matrix equation $U\mathbf{x} = \mathbf{0}$ is equivalent to the statement that $\mathbf{u}_i^* \mathbf{x} = 0$ for all $i$, i.e. $\langle \mathbf{u}_i, \mathbf{x} \rangle = 0$ for all $i$.

**4. Affine Hyperplanes.** An *affine hyperplane* is an affine subspace $H \subseteq V$ of dimension $\dim V - 1$. If $V$ is a real inner product space then one can show using the previous problem that any affine hyperplane $H$ can be expressed as

$$H = H_{\mathbf{u},k} = \{\mathbf{x} \in V : \langle \mathbf{x}, \mathbf{u} \rangle = k\}$$

for some unit vector $\mathbf{u} \in V$ and scalar $k \in \mathbb{R}$. Let $P_{\mathbf{u},k} : V \to V$ denote the orthogonal projection onto $H_{\mathbf{u},k}$ and let $F_{\mathbf{u},k} : V \to V$ denote the orthogonal reflection across $H_{\mathbf{u},k}$.

(a) Fix any point $\mathbf{x}' \in H_{\mathbf{u},k}$, so that $\langle \mathbf{x}', \mathbf{u} \rangle = k$. Show that we can write

$$H_{\mathbf{u},k} = \mathbf{u}^\perp + \mathbf{x}' = \{\mathbf{x} + \mathbf{x}' : \mathbf{x} \in \mathbf{u}^\perp\}.$$

(b) By translating to the origin, one can show that

$$P_{\mathbf{u},k}(\mathbf{x}) = P_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}',$$
$$F_{\mathbf{u},k}(\mathbf{x}) = F_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}'.$$

Use this to find explicit formulas for $P_{\mathbf{u},k}(\mathbf{x})$ and $F_{\mathbf{u},k}(\mathbf{x})$. [Hint: Recall that $P_{\mathbf{u},0}(\mathbf{x}) = \mathbf{x} - \langle \mathbf{u}, \mathbf{x} \rangle \mathbf{u}$ and $F_{\mathbf{u},0}(\mathbf{x}) = \mathbf{x} - 2\langle \mathbf{u}, \mathbf{x} \rangle \mathbf{u}$.]

(c) Compute the projection of the origin onto $H_{\mathbf{u},k}$.

(d) Describe the composition $F_{\mathbf{u},k} \circ F_{\mathbf{u},\ell}$ of two parallel reflections. See the challenge problem for the composition of two non-parallel reflections.

(a): For any point $\mathbf{x}' \in H_{\mathbf{u},k}$, so that $\langle \mathbf{u}, \mathbf{x}' \rangle = k$. Then for any $\mathbf{x} \in V$ we have

$$\begin{aligned}
\mathbf{x} \in H_{\mathbf{u},k} &\iff \langle \mathbf{u}, \mathbf{x} \rangle = k \\
&\iff \langle \mathbf{u}, \mathbf{x} \rangle = \langle \mathbf{u}, \mathbf{x}' \rangle \\
&\iff \langle \mathbf{u}, \mathbf{x} - \mathbf{x}' \rangle = 0 \\
&\iff \mathbf{x} - \mathbf{x}' \in \mathbf{u}^\perp \\
&\iff \mathbf{x} \in \mathbf{u}^\perp + \mathbf{x}'.
\end{aligned}$$

(b): Recall that $\langle \mathbf{u}, \mathbf{x}' \rangle = k$. Using the provided formulas gives

$$\begin{aligned}
P_{\mathbf{u},k}(\mathbf{x}) &= P_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}' \\
&= \big(\mathbf{x} - \mathbf{x}' - \langle \mathbf{u}, \mathbf{x} - \mathbf{x}' \rangle \mathbf{u}\big) + \mathbf{x}' \\
&= \mathbf{x} - \langle \mathbf{u}, \mathbf{x} - \mathbf{x}' \rangle \mathbf{u} \\
&= \mathbf{x} - \big(\langle \mathbf{u}, \mathbf{x} \rangle - \langle \mathbf{u}, \mathbf{x}' \rangle\big)\, \mathbf{u} \\
&= \mathbf{x} - \big(\langle \mathbf{u}, \mathbf{x} \rangle - k\big)\, \mathbf{u},
\end{aligned}$$

and

$$\begin{aligned}
F_{\mathbf{u},k}(\mathbf{x}) &= F_{\mathbf{u},0}(\mathbf{x} - \mathbf{x}') + \mathbf{x}' \\
&= \big(\mathbf{x} - \mathbf{x}' - 2\langle \mathbf{u}, \mathbf{x} - \mathbf{x}' \rangle \mathbf{u}\big) + \mathbf{x}' \\
&= \mathbf{x} - 2\langle \mathbf{u}, \mathbf{x} - \mathbf{x}' \rangle \mathbf{u} \\
&= \mathbf{x} - 2\big(\langle \mathbf{u}, \mathbf{x} \rangle - \langle \mathbf{u}, \mathbf{x}' \rangle\big)\, \mathbf{u} \\
&= \mathbf{x} - 2\big(\langle \mathbf{u}, \mathbf{x} \rangle - k\big)\, \mathbf{u}.
\end{aligned}$$

(c): The projection of $\mathbf{0}$ onto $H_{\mathbf{u},k}$ is

$$P_{\mathbf{u},k}(\mathbf{0}) = \mathbf{0} - \big(\langle \mathbf{u}, \mathbf{0} \rangle - k\big)\, \mathbf{u} = k\mathbf{u}.$$

Note that this is the point on the hyperplane $H_{\mathbf{u},k}$ that is closest to $\mathbf{0}$. We can say that the hyperplane $H_{\mathbf{u},k}$ has signed-distance $k$ from the origin. (Note that $k$ is allowed to be negative when $H_{\mathbf{u},k}$ is "below" the origin with respect to $\mathbf{u}$.)

(d): For any $\mathbf{x} \in V$ we have

$$\begin{aligned}
(F_{\mathbf{u},k} \circ F_{\mathbf{u},\ell})(\mathbf{x}) &= F_{\mathbf{u},k}(F_{\mathbf{u},\ell}(\mathbf{x})) \\
&= F_{\mathbf{u},k}\Big(\mathbf{x} - 2(\langle \mathbf{u}, \mathbf{x}\rangle - \ell)\mathbf{u}\Big) \\
&= \Big(\mathbf{x} - 2(\langle \mathbf{u}, \mathbf{x}\rangle - \ell)\mathbf{u}\Big) \\
&\qquad - 2\bigg(\Big\langle \mathbf{u}, \mathbf{x} - 2(\langle \mathbf{u}, \mathbf{x}\rangle - \ell)\mathbf{u}\Big\rangle - k\bigg)\mathbf{u} \\
&= \Big(\mathbf{x} - 2\langle \mathbf{u}, \mathbf{x}\rangle\mathbf{u} + 2\ell\mathbf{u}\Big) \\
&\qquad - 2\bigg(\langle \mathbf{u}, \mathbf{x}\rangle - 2(\langle \mathbf{u}, \mathbf{x}\rangle - \ell)\langle \mathbf{u}, \mathbf{u}\rangle - k\bigg)\mathbf{u} \\
&= \Big(\mathbf{x} - 2\langle \mathbf{u}, \mathbf{x}\rangle\mathbf{u} + 2\ell\mathbf{u}\Big) \\
&\qquad - 2\bigg(\langle \mathbf{u}, \mathbf{x}\rangle - 2(\langle \mathbf{u}, \mathbf{x}\rangle - \ell) - k\bigg)\mathbf{u} \\
&= \mathbf{x} - 2\langle \mathbf{u}, \mathbf{x}\rangle\mathbf{u} + 2\ell\mathbf{u} - 2\Big(-\langle \mathbf{u}, \mathbf{x}\rangle + 2\ell - k\Big)\mathbf{u} \\
&= \mathbf{x} - 2\langle \mathbf{u}, \mathbf{x}\rangle\mathbf{u} + 2\ell\mathbf{u} + 2\langle \mathbf{u}, \mathbf{x}\rangle\mathbf{u} - 4\ell\mathbf{u} + 2k\mathbf{u} \\
&= \mathbf{x} + 2(k - \ell)\mathbf{u}.
\end{aligned}$$

This is the function that translates every point by the vector $2(k - \ell)\mathbf{u}$. So: The composition of two parallel reflections is a translation.

**5. Left Inverses.** Let $A \in \mathbb{C}^{m\times n}$ have independent columns, so that $m - n \geq 0$. In this problem you will find a bijection between $\mathbb{C}^{n\times(m-n)}$ and the set of left inverses of $A$.

(a) Prove that $\mathcal{N}(A^*A) = \mathcal{N}(A)$. [Hint: Show that $\mathbf{x} \in \mathcal{N}(A^*A)$ implies $\|A\mathbf{x}\| = 0$.]

(b) Use (a) to prove that $A^*A$ is invertible. Show that

$$L := (A^*A)^{-1}A^*$$

is a left inverse of $A$.

(c) Let $N \in \mathbb{C}^{m\times(m-n)}$ have columns forming an orthonormal basis of $\mathcal{N}(A^*)$, and consider any matrix $X \in \mathbb{C}^{n\times(m-n)}$. Define the matrix

$$L_X := L + XN^*.$$

Show that $L_X$ is a left inverse of $A$. [Hint: Show that $N^*A = O$.]

(d) Prove that every left inverse of $A$ has the form $L_X$ for some unique $X \in \mathbb{C}^{n\times(m-n)}$. [Hint: Let $L'$ be any left inverse of $A$ and consider $M = L' - L$. Show that $A^*M^* = O$, so every column of $M^*$ belongs to $\mathcal{N}(A^*)$. It follows that $M^* = NY$ for some matrix $Y$. For uniqueness, use the fact that $N^*N = I$.]

(e) What happens in the case $m = n$?

(a): Note that $A\mathbf{x} = \mathbf{0}$ implies $A^*A\mathbf{x} = A^*\mathbf{0} = \mathbf{0}$, hence $\mathcal{N}(A) \subseteq \mathcal{N}(A^*A)$. For the other direction, suppose that $A^*A\mathbf{x} = \mathbf{0}$. Then we have

$$\|A\mathbf{x}\|^2 = \langle A\mathbf{x}, A\mathbf{x}\rangle = (A\mathbf{x})^*(A\mathbf{x}) = \mathbf{x}^*A^*A\mathbf{x} = \mathbf{x}^*\mathbf{0} = 0.$$

This implies that $\|A\mathbf{x}\| = 0$ and hence $A\mathbf{x} = \mathbf{0}$. We have shown that $A^*A\mathbf{x} = \mathbf{0}$ implies $A\mathbf{x} = \mathbf{0}$, and hence $\mathcal{N}(A^*A) \subseteq \mathcal{N}(A)$.

(b): A square matrix $B$ is invertible if and only if $\mathcal{N}(B) = \{\mathbf{0}\}$. Thus we have

$$\begin{aligned} (A^*A)^{-1} \text{ exists} &\iff \mathcal{N}(A^*A) = \{\mathbf{0}\} \\ &\iff \mathcal{N}(A) = \{\mathbf{0}\} && \text{from part (a)} \\ &\iff A \text{ has independent columns} && \text{from 1(b).} \end{aligned}$$

Since $A$ has independent columns, we conclude that $(A^*A)^{-1}$ exists. Then the matrix $L := (A^*A)^{-1}A^*$ is a left inverse of $A$ because

$$LA = [(A^*A)^{-1}A^*]A = (A^*A)^{-1}(A^*A) = I_n.$$

(c): Let $N$ have columns forming an orthonormal basis of $\mathcal{N}(A^*)$. Since every column $\mathbf{n}_i$ of $N$ satisfies $A^*\mathbf{n}_i = \mathbf{0}$, this implies $A^*N = O_{n\times(m-n)}$, and hence

$$N^*A = (A^*N)^* = O^*_{n\times(m-n)} = O_{(m-n)\times n}.$$

Then for any matrix $X \in \mathbb{C}^{n\times(m-n)}$ we have

$$\begin{aligned} L_XA &= (L + XN^*)A \\ &= LA + XN^*A \\ &= I_n + XO_{(m-n)\times n} \\ &= I_n + O_{n\times n} \\ &= I_n. \end{aligned}$$

(d): Now we show that every left inverse of $A$ has the form $L_X$. Let $L'$ be any left inverse of $A$ and define $M := L' - L$. Note that $MA = L'A - LA = I_n - I_n = O_{n\times n}$, and hence $A^*M^* = (MA)^* = O_{n\times n}$. This implies that every column of $M^*$ belongs to $\mathcal{N}(A^*)$, and hence is a linear combination of the columns of $N$. In other words, $M^* = NY$ for some matrix $Y \in \mathbb{C}^{(m-n)\times n}$. It follows that

$$L' = L + M = L + (NY)^* = L + Y^*N^* = L_{Y^*},$$

and we can take $X = Y^*$. To prove uniqueness, suppose that $L_{X_1} = L_{X_2}$, so that

$$\begin{aligned} L + X_1N^* &= L + X_2N^* \\ X_1N^* &= X_2N^*. \end{aligned}$$

Since the columns of $N$ are orthonormal, we have $N^*N = I_{m-n}$, hence we can multiply both sides on the right by $N$ to obtain $X_1 = X_2$.

(e): If $A$ is a square matrix with independent columns, then $A^{-1}$ exists, which implies that $A^{-1}$ is the unique left inverse of $A$. Note that

$$L = (A^*A)^{-1}A^* = A^{-1}(A^*)^{-1}A^* = A^{-1},$$

so our formula still works in this case. If we interpret $N^*$ as the unique matrix of shape $0 \times n$ and $X$ as the unique matrix of shape $n \times 0$, then we could say that $XN^* = O_{n\times n}$, and hence $A^{-1} = L + XN^*$, but maybe that's too pedantic.

**Problem 6. Least Squares.** Consider four data points:

$$(x, y) = (1, 1), (2, 1), (3, 3), (4, 5).$$

(a) Find the OLS best fit line $y = mx + b$ for these points. Draw your answer.
(b) Find the OLS best fit parabola $y = ax^2 + bx + c$ for the same points. Draw your answer.

(a): Each data point gives a linear equation in $m$ and $b$. This system of 4 linear equations in 2 unknowns has no solution, so we solve the normal equation:

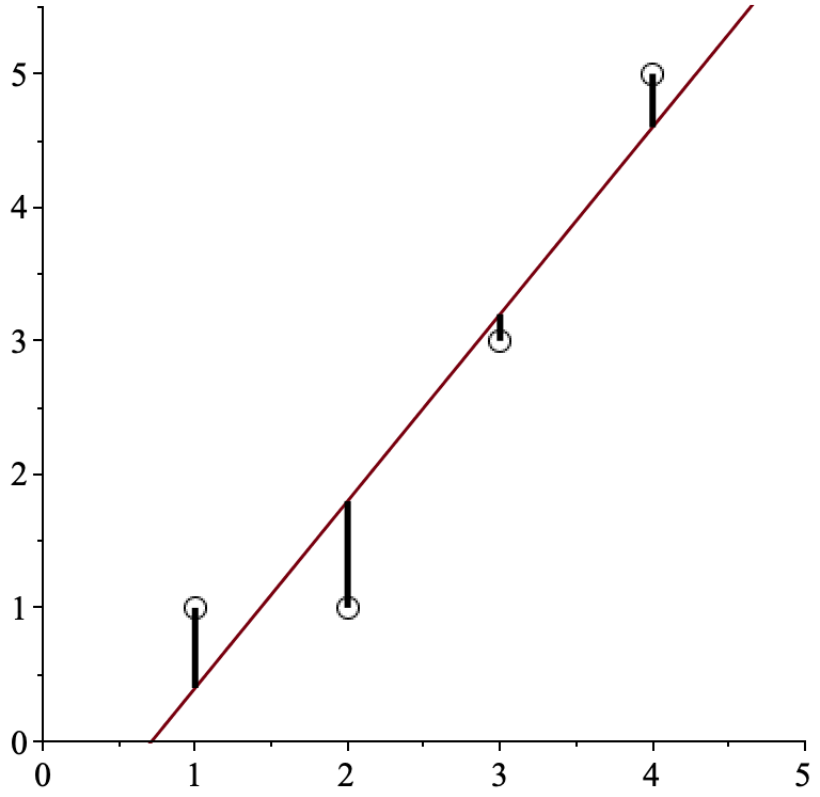
$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} \hat{m} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix} \begin{pmatrix} \hat{m} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 32 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} \hat{m} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 7/5 \\ -1 \end{pmatrix}.$$

The best fit line is $y = \hat{m}x + \hat{b} = (7/5)x - 1$. Here is a picture:



(b): Each data point gives a linear equation in $a, b, c$. This system of 4 linear equations in 3 unknowns has no solution, so we solve the normal equation:

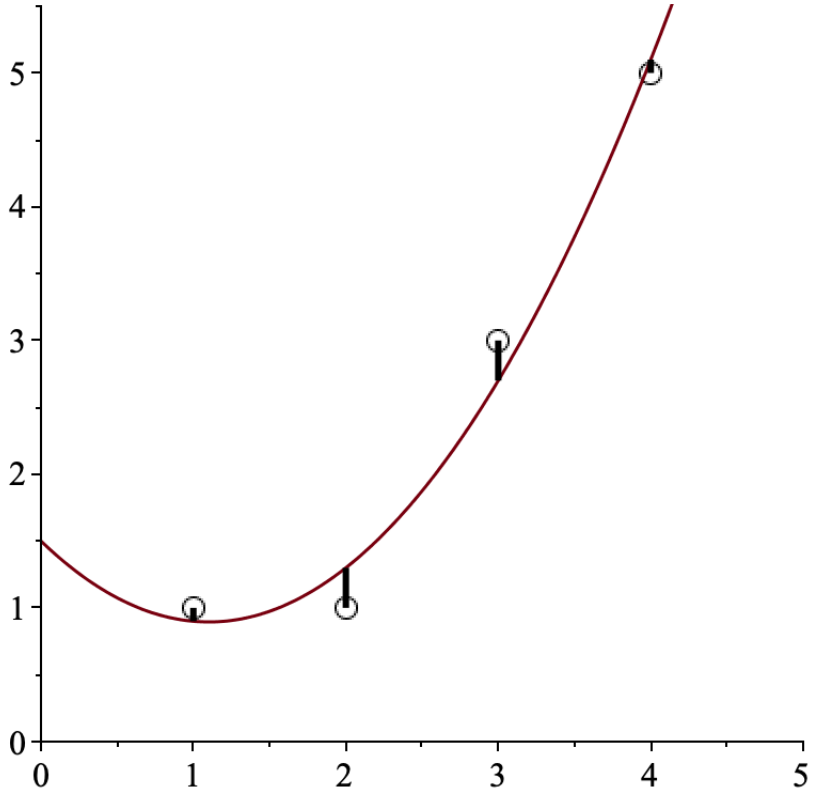
$$\begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 9 & 16 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{pmatrix} = \begin{pmatrix} 1 & 4 & 9 & 16 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 354 & 100 & 30 \\ 100 & 30 & 10 \\ 30 & 10 & 4 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{pmatrix} = \begin{pmatrix} 112 \\ 32 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{pmatrix} = \begin{pmatrix} 1/2 \\ -11/10 \\ 3/2 \end{pmatrix}$$

The best fit parabola is $y = \hat{a}x^2 + \hat{b}x + \hat{c} = (1/2)x^2 - (11/10)x + (3/2)$. Here is a picture:




Observe that this parabola is a "better fit" than the best fit line. That is, the sum of the squares of the vertical errors is smaller. In fact, one can check that these sums are 6/5 in (a) and 1/5 in (b). So I guess (b) is six times "better" than (a). For degrees $d \geq 3$ there exists a curve of degree $d$ passing through all four data points, so in these cases the error is zero.

**Challenge Problem (Euler-Rodrigues Formula).** For any unit vector $\mathbf{u} \in \mathbb{R}^n$, recall that $F_{\mathbf{u}} = I - 2\mathbf{u}\mathbf{u}^T$ is the matrix that reflects across the hyperplane $\mathbf{u}^\perp$. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be unit vectors with angle $\alpha$ between them (assume $0 < \alpha < \pi$) and let $U = \text{Span}\{\mathbf{u}, \mathbf{v}\} \subseteq \mathbb{R}^n$ be the plane that they span. We will show that the product matrix $R := F_{\mathbf{u}} F_{\mathbf{v}}$ fixes $U^\perp$ and rotates $U$ by angle $2\alpha$. To simplify notation we write $c := \cos\alpha$ and $s := \sin\alpha$. The key idea of the proof is to consider the matrix

$$J := \frac{1}{s}(\mathbf{u}\mathbf{v}^T - \mathbf{v}\mathbf{u}^T).$$

We will see that $J$ vanishes on $U^\perp$ and performs a quarter turn on $U$.

(a) Show that $\mathbf{x} \in U^\perp$ implies $J\mathbf{x} = \mathbf{0}$.

(b) For $\mathbf{x} \in U$, show that $J^2\mathbf{x} = -\mathbf{x}$. [Hint: Show that $J^2\mathbf{u} = -\mathbf{u}$ and $J^2\mathbf{v} = -\mathbf{v}$.]

(c) For $\mathbf{x} \in U$, show that $J\mathbf{x} \in U$, $\|J\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle J\mathbf{x}, \mathbf{x}\rangle = 0$. In other words, $J$ rotates the plane $U$ by 90°. [Hint: Use (b) and the fact that $J^T = -J$.]

(d) Show that

$$\boxed{R = I + \sin(2\alpha)J + (1 - \cos(2\alpha))J^2}.$$

This is called the *Euler-Rodrigues formula* for rotation matrices. [Hint: Use the trig identities $\sin(2\alpha) = 2cs$ and $1 - \cos(2\alpha) = 2s^2$.]

(e) For $\mathbf{x} \in U$, show that

$$R\mathbf{x} = \cos(2\alpha)\mathbf{x} + \sin(2\alpha)J\mathbf{x}.$$

Explain why this shows that $R$ rotates the plane $U$ by angle $2\alpha$.

(f) For any square matrix, one can define the *matrix exponential*

$$\exp(A) := I + A + \frac{1}{2}A^2 + \frac{1}{6}A^3 + \cdots = \sum_{k \geq 0} \frac{1}{k!}A^k,$$

which converges with respect to any matrix norm. Use the Euler-Rodrigues formula to show that

$$\boxed{R = \exp(2\alpha J)}.$$

The matrix $J$ is called an *infinitesimal rotation*. [Hint: Parts (a) and (b) imply that $J^3 = -J$, so that $J^{2r+1} = (-1)^r J$ and $J^{2r+2} = (-1)^r J^2$ for all integers $r \geq 0$. Use the Taylor series for sin and cos.]

Sorry, I don't have time to type up the solution.